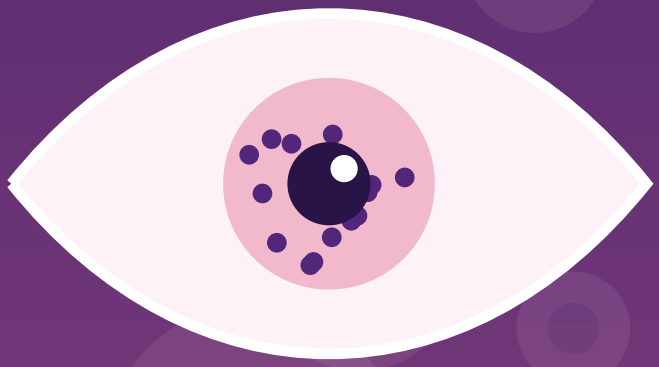


littlEye AI

Manual for physicians and students

version 4.0 · free · iPhone, iPad, Mac

Training and evaluating AI models from histology slides —
entirely on-device, nothing in the cloud.

Forensika

OBJASŇUJEME · SPÁJAME · POSÚVAME

forensika.eu

Ján Šikuta, MD, PhD · Comenius University Bratislava · app 4.0 (24) · October 2026

Contents

Where to find what

1 What littlEye AI is and who it is for
the idea · free · privacy
2 What is new in 4.0
navigation · Dataset · instruments · backup · shortcuts
3 First launch
language, name, sample model
4 Navigation and the Overview
sidebar · iPhone tabs · Next step · Project
5 Dataset: images and diagnoses
grid of 27 organs, import, drag & drop, crop
6 Slide review by colleagues
digits 1–9, undo last vote, majority decides
7 Training a model
readiness traffic light, profiles, time estimate
8 The training result and what to do with it
quality sentence, confusion matrix, what to add
9 Evaluating an image
model chaining, field-of-view crop, preliminary diagnosis
10 The learning loop
learn from a correction, when to retrain
11 Cases
slides per case, assessment
12 Sharing: .le3AI models and .le3set datasets
what travels, 3.x compatibility, encryption
13 Importing a model trained outside the app
.mlpackage / .mlmodel, provenance, in-app test
14 Atlas, Education, Methodology, References
103 diagnoses, stains, IHC, citations
15 Image quantification and desmin IHC
instrument panel · recipes · study mode · CPA and indices
16 The measurement log
table, batches, ICC and Bland–Altman, CSV export
17 Settings, backup and restore
four tabs · complete backup · restore from backup
18 littlEye AI on the Mac
menus, shortcuts, windows, drag & drop
19 Privacy, limits, collaboration and support
what the app does not do · collaboration · voluntary contribution

How to read this manual

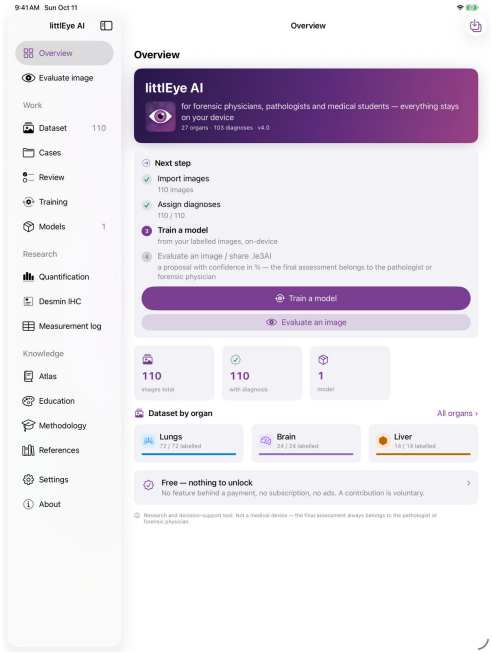
Chapters 3 – 8 are the path from an empty app to your first own model — read them in order. The other chapters are for reference. Changes since 3.x are summarised in chapter 2, navigation in chapter 4. Screenshots are from Mac, iPad and iPhone; the Mac has its own chapter 18 with all shortcuts, the image-analysis tools chapter 15.

1 What littlEye AI is and who it is for

the idea · free · privacy

littlEye AI grew from a simple idea: the microscopic world gives meaning to the macroscopic one. It is built for forensic pathologists and pathologists in practice, and above all for medical students as a companion to histology and pathology.

The app does three things: it collects H&E; microphotographs with diagnoses, trains a small model from them right on your iPad, iPhone or Mac, and evaluates new images — always with a confidence percentage, never without one. On top of that it adds a knowledge base: a diagnosis atlas, an education module on stains and immunohistochemistry, and the list of sources the content is based on.



Overview on the iPad: the Next step card, image and model counts, Dataset by organ and the Project card.

Free — and why

Since version 3.0 littlEye AI is free and fully functional. No subscription, no feature locked behind a payment, no advertising. The people it was built for — students and junior colleagues — are the same people least able to afford it; and the point of the project is that departments exchange trained models. That needs people, not revenue. Those who wish can make a voluntary contribution (chapter 19). It unlocks nothing.

Privacy as a principle

The entire app works on-device. Images, cases and models never leave it, and the app makes no network connections of its own — the only exceptions are the voluntary contribution through the App Store and literature links, which open in the browser. For forensic medicine this is not an extra feature but the condition without which the app could not exist.

> **What littlEye AI is not**
>
> It is not a medical device and does not give a diagnosis. It is a research and decision-support tool: the model suggests, the physician decides. That sentence is on every content screen of the app on purpose.

2 What is new in 4.0

navigation · Dataset · instruments · backup · shortcuts

Version 4.0 brings a new look and simpler work on Mac, iPad and iPhone. The .le3AI and .le3set files remain fully compatible with version 3.x in both directions — a model or dataset from the previous version opens without changes, and a file from 4.0 also opens in an older littlEye AI.

Clearer navigation

- The sidebar has 15 items in the order you work: Overview · Evaluate image · Work (Dataset, Cases, Review, Training, Models) · Research (Quantification, Desmin IHC, Measurement log) · Knowledge (Atlas, Education, Methodology, References) · About (chapter 4).
- New Dataset section: all 27 organs as a grid grouped by organ system, with image and labelled counts (chapter 5).
- iPhone has five tabs: Overview · Evaluate · Dataset · Atlas · More.
- Overview suggests the next step (import → assign diagnoses → review → train → evaluate and share). Collaboration and support reminders no longer sit above the working sections but in the Project card on the Overview.
- An image in progress, a result or a training run is not lost when you switch sections; training continues and the sidebar shows its progress.

Research tools as an instrument

- Quantification and Desmin IHC: the image and a large result with its interval side by side, settings and verification (fields, point counting, sensitivity) below; study mode as a status strip (chapter 15).
- Desmin calculates live, without an Evaluate button. Index 2 has a single path — the DAB + H&E image pair; Quantification sends the image to Desmin, not the numbers.
- The measurement engine is 5 to 10 times faster and the threshold slider responds instantly.
- The Measurement log is its own section: on the Mac a sortable table, deletion of single entries and CSV export via Save As (chapter 16).

Data and sharing

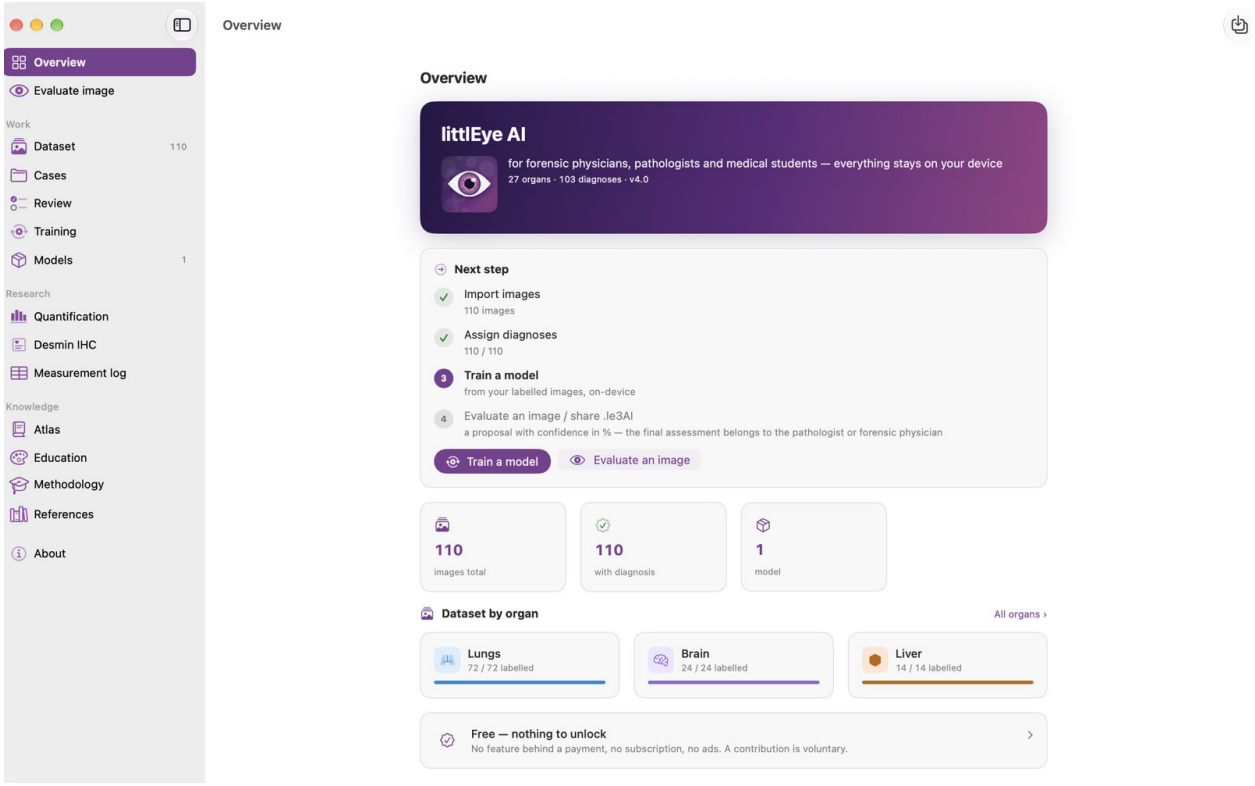
- A complete backup including the measurement log and staining batches, and a new restore from backup — in Settings (chapter 17).
- Reviews in .le3set files are matched through the diagnosis catalogue, images are not re-encoded on import (no quality loss) and exporting a model keeps its original author.
- Slide review: digits 1–9, Space skips, ⌘Z undoes the last vote; vote ties are resolved deterministically.
- The test set is split by case so the test result is not inflated. A damaged archive index is no longer overwritten — the app offers to restore the previous version.

Mac

- Settings (⌘,) with four tabs — General, Appearance, Reminders, Backup — and a native About window; new shortcuts ⇧⌘A, ⇧⌘H, ⇧⌘L, ⇧⌘I, ⇧⌘N, ⇧⌘E and ⌥⌘A; ⌘1–⌘0 stay (chapter 18). Context menus and an inspector in Models.

Design and accessibility

- A unified design derived from the app icon, a full dark mode and colours with verified contrast. VoiceOver labels for controls and charts, Dynamic Type and numbers formatted for the app language.

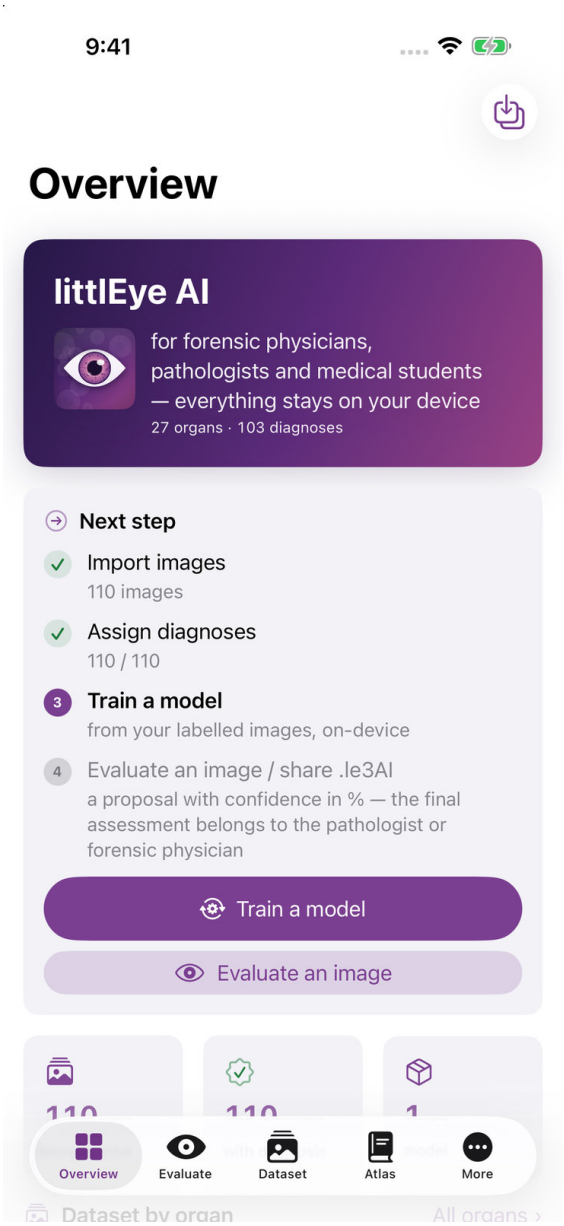


Mac, version 4.0: the sidebar in the order you work and the Overview with the Next step card.

3 First launch

language, name, sample model

1. On first launch the welcome screen appears. Choose the language (Slovak / English) and enter the reviewer name. The name is stored with every slide review and in every exported model — an audit trail, not a registration. It is not sent anywhere.
2. Tap Try with the sample model: DEMO has 10 diagnoses trained on synthetic patterns — not real slides — and the app takes you straight to Evaluate image. Or choose Start with my own slides and you go to Dataset.
3. From now on the Next step card on the Overview guides you (chapter 4): it shows what makes sense to do next, and its button leads straight there.
4. Switch Professional / Lay terminology and the language in Settings. Professional shows diagnoses professionally (“steatosis”), lay plainly (“fatty liver”). Exports and training always use professional keys, so the switch breaks nothing.
5. You can reopen the welcome screen from About.



Tip for departments

Set a real reviewer name on every device — in colleague review the majority of reviews decides, and the exported model will show who trained it.

4 Navigation and the Overview

sidebar · iPhone tabs · Next step · Project

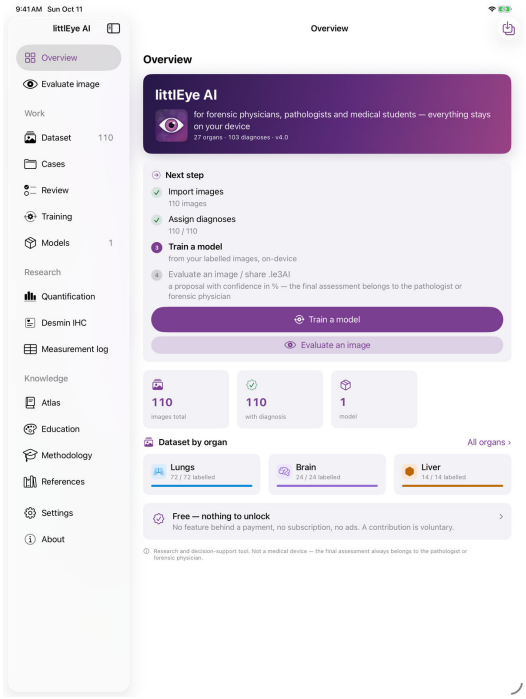
On Mac, iPad and iPhone the app has the same sections; only the way you move between them differs. Every section has a title, one sentence on what it does, a main action and, at the end, always the note that this is a research and decision-support tool, not a medical device.

The sidebar (Mac and iPad)

The sidebar has 15 rows in the order you work. Overview and Evaluate image sit at the top without a heading; then three groups:

	Section	What it is for
Top	Overview · Evaluate image	Next step and project status · evaluating a single image
Work	Dataset · Cases · Review · Training · Models	collect images, review with colleagues, train, manage models
Research	Quantification · Desmin IHC · Measurement log	measure colour and shape on an image, record measurements
Knowledge	Atlas · Education · Methodology · References	diagnoses, stains, methodology and literature
Bottom	Info	about the project, privacy, collaboration, support

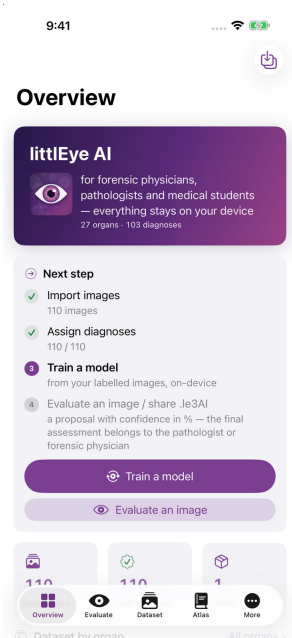
- On the right of a row are small figures: the number of images in Dataset, the number of models, entries in the Measurement log, unreviewed slides in Review and, during training, progress in percent.
- Settings is not a sidebar row on the Mac — you open it with ⌘, in a separate window with four tabs. On the iPad it is a sidebar row, on the iPhone an item under More (chapter 17).
- The sidebar remembers the last section and the open organ after closing the window and restarting.
- ⌘1–⌘9 and ⌘0 lead to sections just as in version 3.x; the order in the sidebar follows the workflow, not the numbers. All shortcuts are in chapter 18.



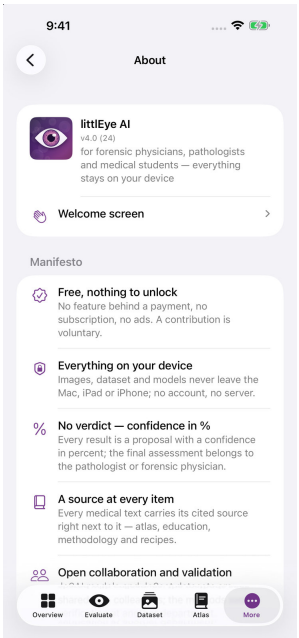
iPad: the same sidebar and the Overview with the Next step card.

iPhone: five tabs

On the iPhone you stand at the microscope — you photograph, add to the dataset, evaluate an image and browse the atlas. Hence the tabs at the bottom: Overview · Evaluate · Dataset · Atlas · More. Everything else is under More, divided into the same groups as in the sidebar (Work: Cases, Review, Training, Models · Research: Quantification, Desmin IHC, Measurement log · Knowledge: Education, Methodology, References · App: Settings, About). Each tab has its own history, so going back always returns you to where you were.



iPhone: Overview.



iPhone: More — the other sections.

The Overview

The Overview consists of cards from top to bottom:

- A header with the icon, the name, one sentence about the app and the number of organs and diagnoses.
- Next step — five steps: Import images → Assign diagnoses → Review with colleagues (optional; shown when at least two people review) → Train a model → Evaluate an image / share .le3AI. The current step has a button that takes you to the right place: with no images “Load sample model” and “Add images”, with unlabelled images Review, with labelled images and no model Training, after ten new images “Retrain”, otherwise “Evaluate image” and “Export .le3AI”. When everything is done the card collapses into a line “Everything is ready”.
- Three figures: images, images with a diagnosis, models. Below them the Dataset by organ card with the six fullest organs and an “All organs” link.
- Project — the only place where the app asks for anything: the collaboration and validation offer (shown only twice), the voluntary-contribution reminder (from the third launch at the earliest, with a “Not now” button) and a permanent line “Free — nothing to unlock”. Reminders can be switched off in Settings → Reminders.
- Notice: a research and decision-support tool, not a medical device.

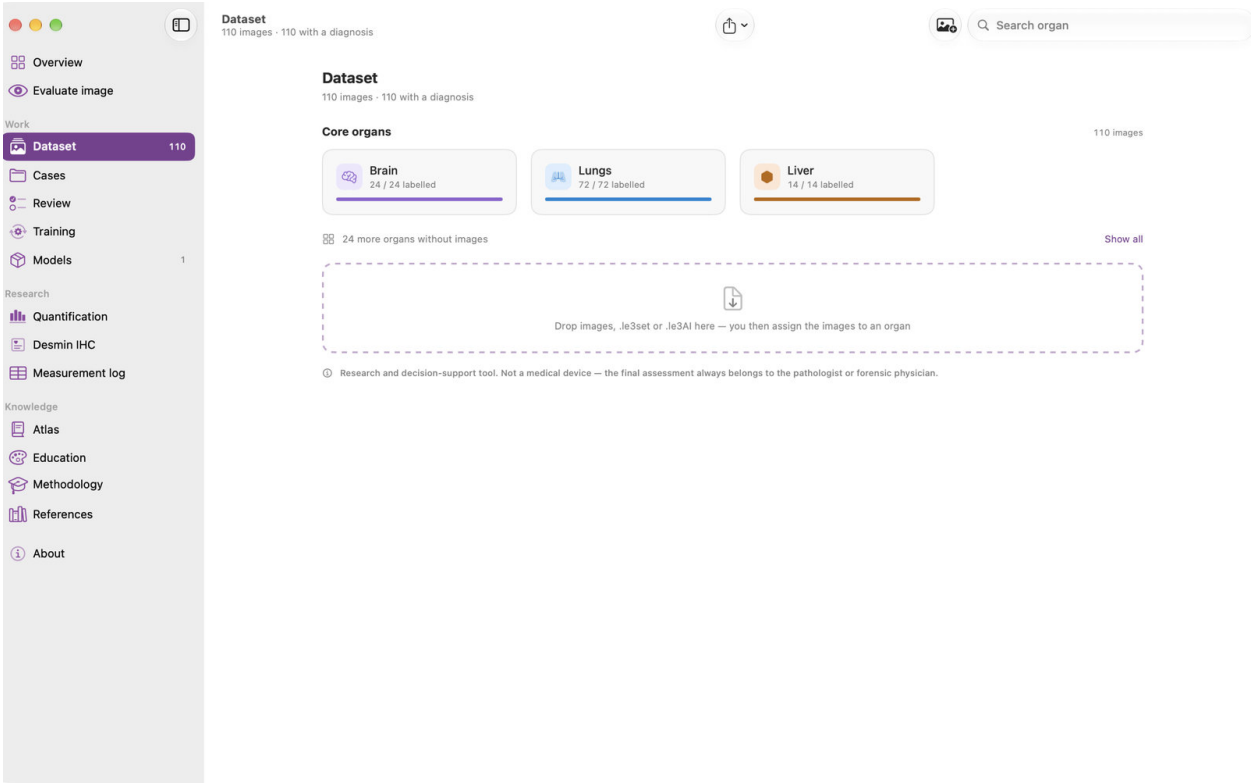
What is not lost when you switch

The app, not the screen, holds the state of evaluation, training and the measurements in Quantification and Desmin. You can start training, go to the Atlas and come back — training keeps running, the sidebar shows its percentage and the image in Evaluate is still loaded.

5 Dataset: images and diagnoses

grid of 27 organs, import, drag & drop, crop

The dataset is organised by organ: 27 organs and tissues in seven systems. Each image has an organ, optionally a diagnosis, magnification and stain. The model learns from the whole field of view — so the finding should fill a substantial part of the image.

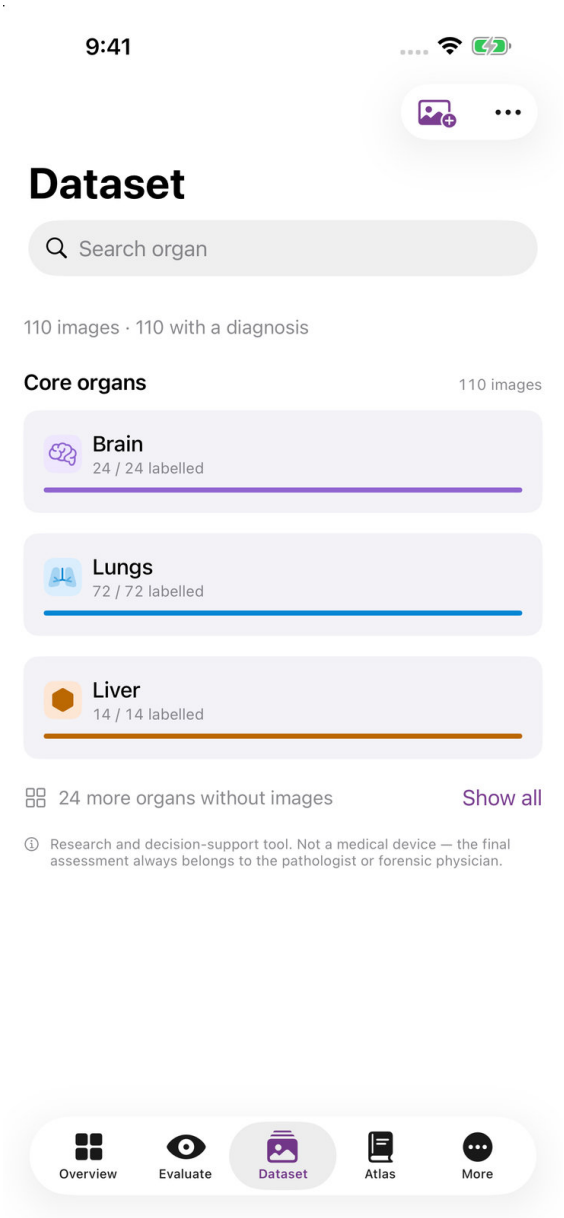


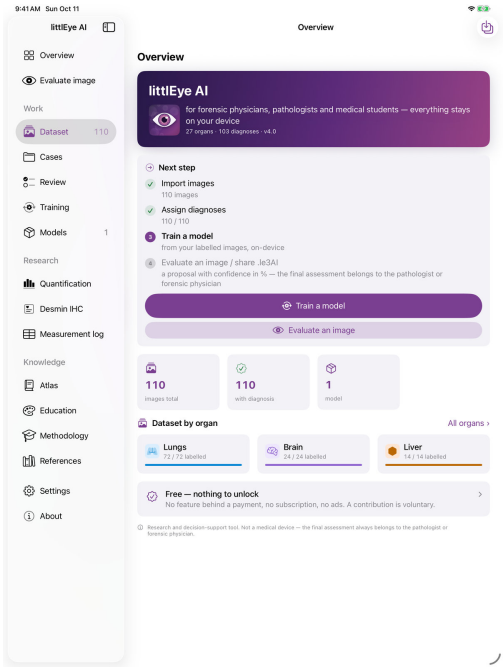
Dataset on the Mac: the grid of organs by system; each tile shows the number of images and how many are labelled.

The organ grid

- Each of the 27 organs is a tile with an icon, a name, the number of images and an “N / M labelled” indicator. The tiles are grouped by organ system.
- Organs without images are dimmed and collapsed under the line “N more organs without images — Show all”. You can search by organ name.
- Clicking (tapping) a tile opens the organ: the grid of its images, the diagnosis filter, import and the magnification and stain settings. The Dataset button takes you back.
- .le3set and .le3AI files and images can be dropped straight onto an organ tile (Mac, iPad); the images are imported into that organ. When you drop onto the window outside an organ, the app asks which organ the images belong to.

1. In Dataset choose an organ. In its toolbar set the import magnification and stain — they are recorded with every image and the model remembers them later. The defaults are in Settings.
2. Press Add images (⇧⌘I) and import images from Photos or Files (JPEG, PNG, TIFF, HEIC). Larger batches import in the background with “N / M” in the toolbar. Large scans are normalised to 2048 px; images are not re-encoded on import, so they lose no quality.
3. Tap an image to open its details: zoom, diagnosis, note, reviews. On the Mac a click shows the inspector on the right, a double-click opens the image in a separate window. A right-click offers assigning a diagnosis, moving to a case and deleting.
4. If an image contains several findings, use a crop: in the details drag to mark the finding and save it as a separate training image. That is how you tell the model where to look.
5. A catalogue diagnosis in an organ can be hidden, a custom one deleted (with confirmation). Deleting an image and batch deletion both ask for confirmation.





iPad: Dataset in the sidebar; on the iPhone (above) the organs are in two columns.

> **The most common trap: the model learns the stain, not the tissue**
>
> If a whole class comes from one slide, the model does not learn the tissue but the tint of that slide — and fails on a slide from another laboratory. Every class should have images from at least three different cases or blocks, ideally from different staining runs too. Training shows you this with a colour traffic light (chapter 7).

Capture rules

- One magnification per model (100× or 200× recommended) — do not mix.
- Sharp, evenly exposed images with correct white balance.
- Do not photograph section edges, folds, creases, bubbles or staining artefacts.
- Several images per slide from different representative places; the finding should fill most of the field of view.
- “Normal” is a full class — the model needs it as much as the pathologies.

6 Slide review by colleagues

digits 1–9, undo last vote, majority decides

Anyone can assign a diagnosis directly in the image details. For joint work, Review is better: each colleague goes through the images under their own name and the final diagnosis is set by the majority of reviews. Every review is stored with name and date.



Review on the Mac: the image with diagnoses numbered 1–9, the organ filter and the review progress.

1. Open Review (⇧⌘H), choose an organ (the default is the one with the most waiting images) and switch on Unreviewed only if you wish. The reviewer name is shown in the toolbar; change it in Settings.
2. Pick a diagnosis by clicking or with a digit 1–9. The vote is saved and the app moves on to the next image; the confirmation “Saved · 12/48” appears at once. If you disagree with colleagues a notice is shown.
3. If unsure, skip the image (Space). A wrong label hurts more than a missing one. The ← → arrows move between slides and ⌘Z undoes the last vote.
4. The question mark beside a diagnosis opens its Atlas description — this unifies criteria across the team. A tie of votes is always resolved the same way, so two colleagues get the same result.
5. A manual assignment in the image details overrides the majority — use it for the final decision when reviews disagree.

Example from practice

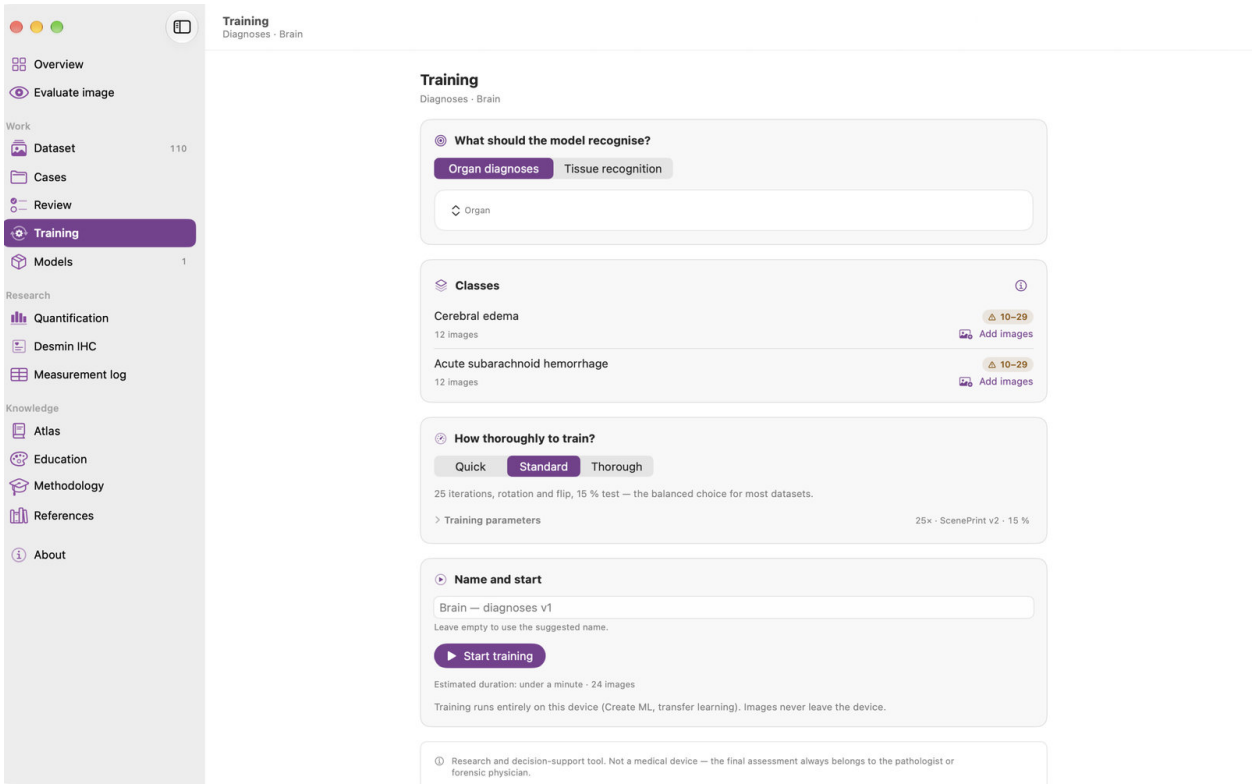
Three colleagues review 40 lung images. They disagree on six (oedema vs. congestion). The majority sets the diagnosis; the team looks at two contested images together and decides manually in the details. Only then do they go into training.

7 Training a model

readiness traffic light, profiles, time estimate

Training runs entirely on-device (Create ML, transfer learning over the ScenePrint extractor). It is built so that you need not know terms like holdout or iteration: the app tells you whether the dataset is ready and offers three profiles.

1. Choose the type: Organ diagnoses (a single-organ model) or Tissue recognition (a model that tells organs apart). Organs are split into “ready” and “not enough images yet”.
2. The Classes and image counts card shows a traffic light: green 30+ images from several cases, yellow 10–29, red fewer than 10. Below it is a concrete hint — “Add 12 images of Normal — ideally from another case” — and a button straight into the organ.
3. Pick a profile: Quick (in a minute you know whether the data works), Standard (for most datasets) or Thorough (the most honest estimate, takes longest). Advanced parameters stay below for those who want them.
4. The model name is prefilled (“Lungs — diagnoses v2”) — the version number grows by itself. Press Start training. The duration estimate is under the traffic light; training can be cancelled any time. While it runs you see the phases (preparation, extraction, training, test, saving) and the elapsed time; you can switch sections — training keeps running and the sidebar shows its percentage.



Training on the Mac: the cards What should the model recognise, Classes with the traffic light, profile and start; after finishing, the result is placed above the form.

What happens under the hood

A portion of images (10–20 % by profile) is set aside as an independent test set — the model never sees them during training and accuracy on them is the most honest estimate of performance. If the smallest class has fewer than 20 images after the holdout, the app also disables Create ML's internal validation:

with 1–2 images it would show 0 % or 100 % and mean nothing.

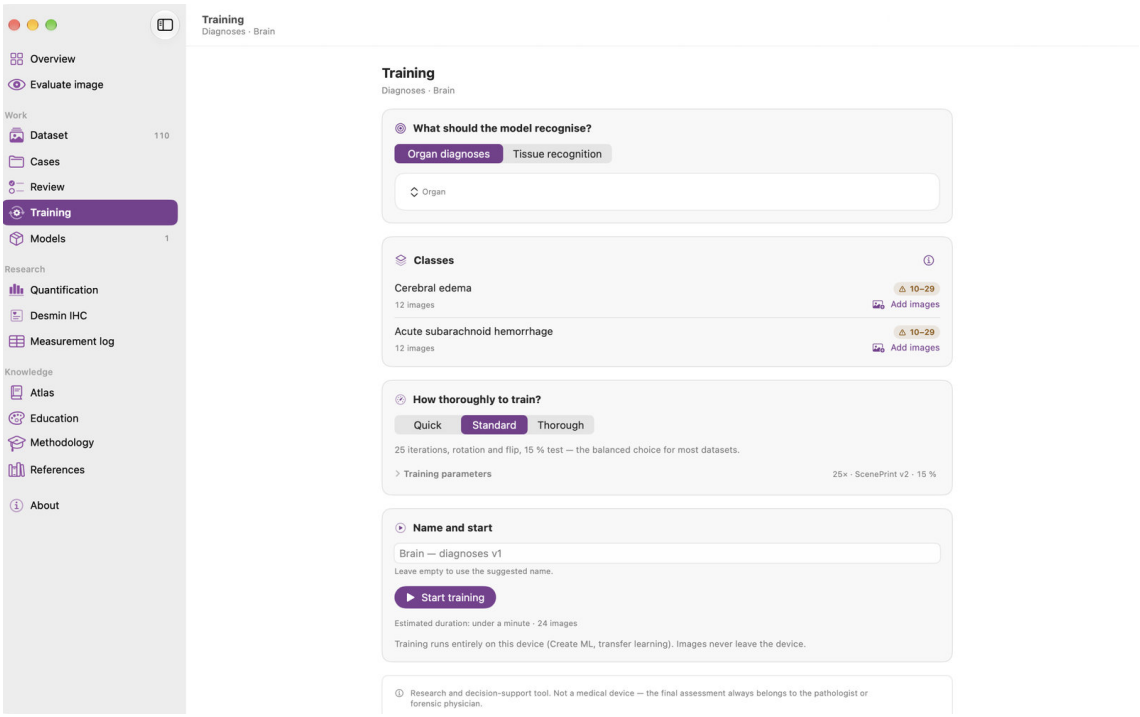
Minimum and recommendation

The minimum is 2 classes of 5 images — enough to try the path, not to make a model. A reasonable start is 30+ images per class from several cases, balanced between classes. A model with 112 normal images and 3 infarct images will answer “normal” to everything.

8 The training result and what to do with it

quality sentence, confusion matrix, what to add

After training you get a card with one sentence about quality, the metrics and the confusion matrix. The sentence is derived from the numbers by rules — no further AI — and says what to do with the model.



The result card is placed above the training form: quality sentence, training / validation / test, confusion matrix and the buttons Evaluate image, What to add, Export .le3AI.

What the sentences mean

- Usable for triage (test ≥ 90 %) — the model can pre-sort images; the final assessment remains with the physician.
- Indicative (75–90 %) — good as a second opinion, not for triage.
- Unsure about class X — confuses it with Y — the confusion matrix showed a specific weakness; the button What to add to improve it lists which classes need images.
- Overfitted (training 20+ % better than test) — the model memorised images instead of tissue. Variety (other cases) helps, not more iterations.
- The test set is too small — with a handful of test images the number proves nothing. Add images and retrain.

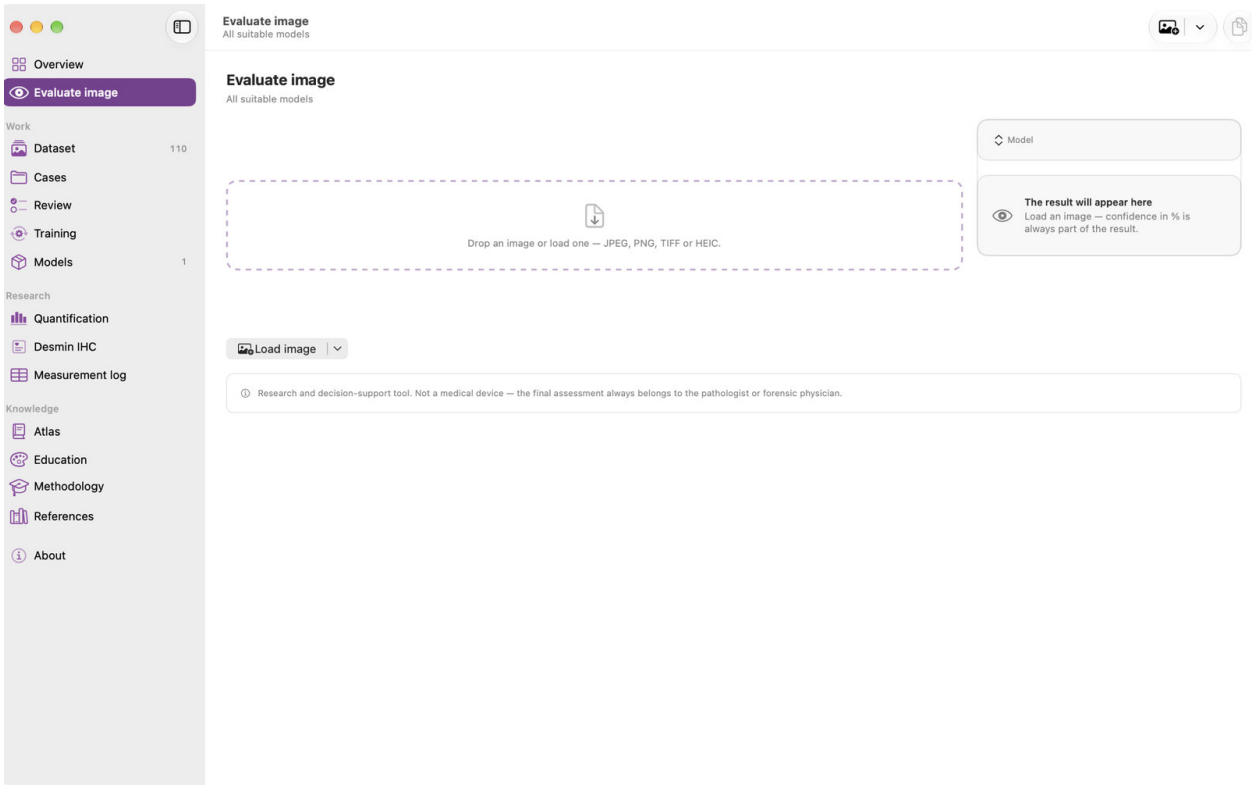
The confusion matrix

Row = actual diagnosis, column = what the model said. Numbers on the diagonal are correct answers. If the “Oedema” row has many in the “Congestion” column, the model confuses those two — and exactly those two classes need more varied images. Below the matrix is each class's sensitivity and precision.

The model is stored inside the app — you will find it in Models. A .le3AI file is created only when you export (⇧⌘E; chapter 12). The button Evaluate image takes you straight to evaluation with this model.

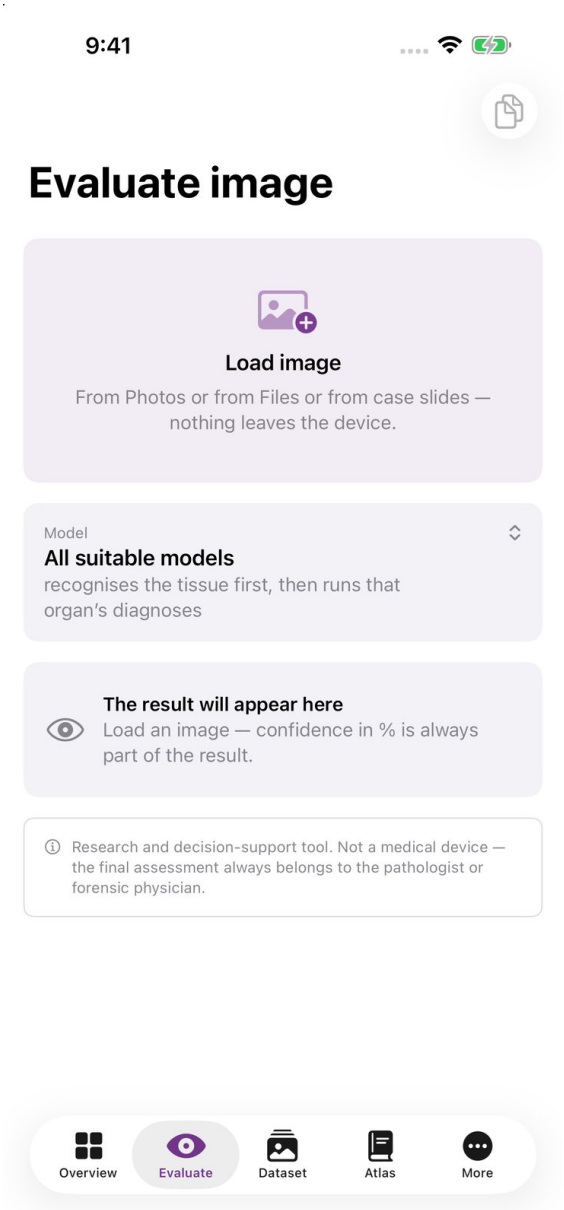
9 Evaluating an image

model chaining, field-of-view crop, preliminary diagnosis



Evaluate image on the Mac: the image with a crop on the left, the result with confidence in percent on the right and the preliminary diagnosis below it.

1. Open Evaluate image (⌘3). The default All suitable models first recognises the tissue and then runs only the diagnostic models of that organ; models of other organs are deliberately skipped and the app says why.
2. Add an image from Photos, from Files, or drop it onto the image area (iPad, Mac). Above the image you see what the model was trained on (“100× · H&E”) — the key fact when tuning inaccurate results.
3. For an overview image use Select field of view (crop) and mark a patch the size of one field of view. Only the crop is evaluated — structures come to the scale the model was trained on. This alone fixes most nonsensical results.
4. The result always carries a confidence percentage. For the most confident class the app offers a preliminary diagnosis: microscopic description and a suggested conclusion from the knowledge base, with a Copy report draft button and source citations.



How to get the most accurate result from a poor image

The model does not compare the image with a textbook — it compares it with what it saw in training. Match the scale with a crop, remove from the field everything that is not the finding, evaluate 3–5 different fields from the same slide: a consistent answer is trustworthy, scattered answers mean the image is outside what the model knows. And check the class list — the model only ever chooses among the classes it learned.

10 The learning loop

learn from a correction, when to retrain

Core ML models cannot be fine-tuned — they are “baked”. Learning therefore goes through the dataset: you save the corrected image with the right diagnosis, and at the next retraining a new model is created that already knows it.

- In Evaluate image, below the result, is the Teach card with the button Save to dataset with diagnosis…: pick the correct organ and diagnosis (the recognised organ is prefilled) and the image is saved into the dataset.
- The app counts how many labelled images were added since each model was trained. At 10 or more, the Overview (Next step card) and Models show the prompt “N images added since training — retrain”.
- Retraining creates a model with the next version number; the old one stays until you delete it — so you can compare whether the new one really helped.

Example

A liver model labels steatosis as normal. In the Teach card the physician picks “Steatosis” and saves the image. After ten such corrections the app suggests retraining; the new v3 model has 15 % higher sensitivity for steatosis — visible in the confusion matrix.

11 Cases

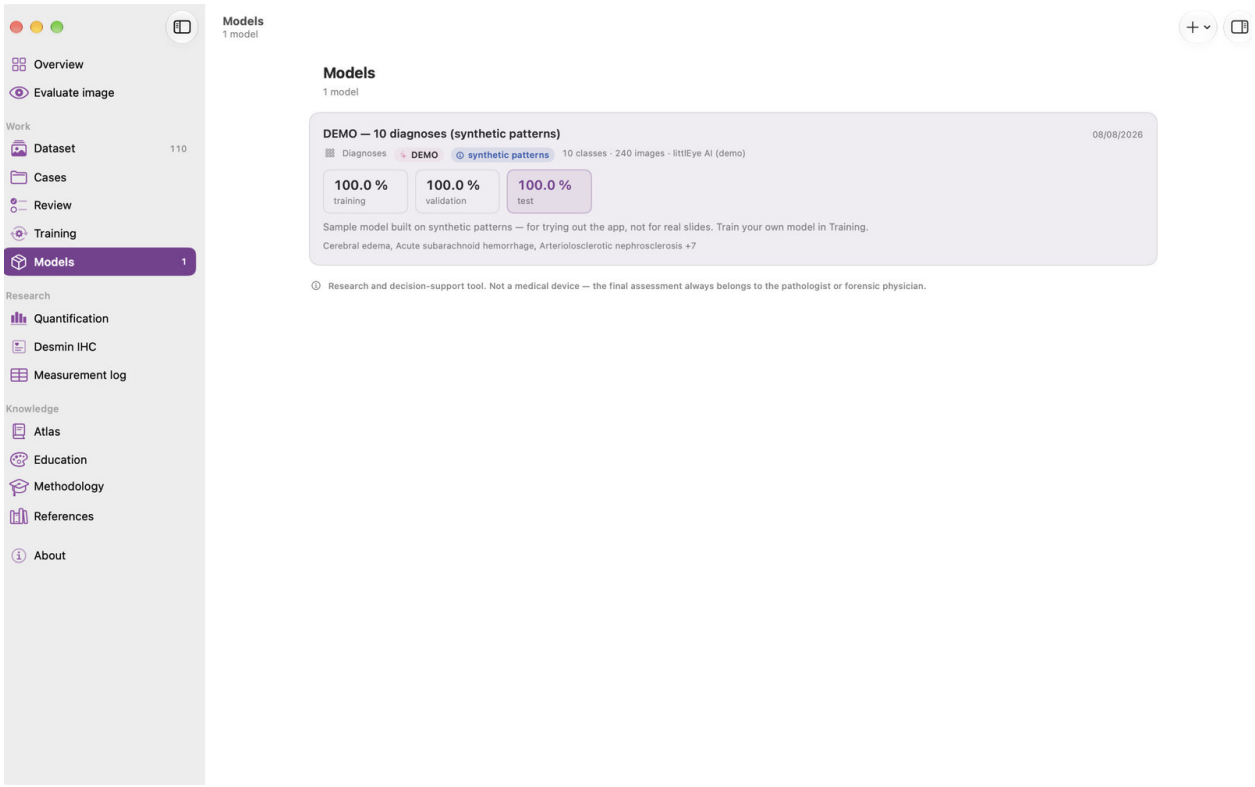
slides per case, assessment

1. Cases (Work group) group scanner slides under a protocol number. Case slides with a diagnosis also count towards the training dataset. Create a new case with the button or ⇧⌘N; the list has a context menu (Open, Archive, Delete…).
2. Add context to the case: history, autopsy conclusion, toxicology. They are inputs for the assessment and are not sent anywhere.
3. Assess case evaluates all slides (physician diagnoses + available models) and generates a concise summary: findings by organ, a “slides only” and a “with case data” conclusion. The Copy button pastes it into Word.
4. If you have a tissue-recognition model, every slide is checked with it: on a mismatch with ≥ 85 % confidence the summary gets an Attention — tissue check section (suspected slide mix-up).
5. A complete library backup (images, cases, models, measurement log) is made in Settings → Backup; the same command is in the File menu. A backup is always complete, never incremental.

12 Sharing: .le3AI models and .le3set datasets

what travels, 3.x compatibility, encryption

Collective work rests on two files. .le3AI is a model: one file with the model, metrics, confusion matrix, magnification, stain, data provenance and licence. .le3set is a dataset: images with diagnoses and reviews — without any case or patient data.



Models on the Mac: every model with metrics and the quality sentence; a right-click offers Details, Export .le3AI, Test on my images, Delete. The details open in the inspector on the right.

Why images are shared, not just models

A model cannot be extended with more images. If a department wants to combine material from three hospitals, each sends its .le3set, the recipient merges them (re-importing the same material does not duplicate it) and retrains the model from scratch on the larger whole. The finished model then goes back as .le3AI.

- Export: Models → Export .le3AI… (⇧⌘E); dataset: organ → toolbar → Sharing → Export .le3set… (respects the diagnosis filter; a “What is shared” preview appears before the export).
- Import: double-click the file, AirDrop, Files, or ⌘O on the Mac / iPad with a keyboard. Images from a .le3set sort themselves into organs.
- Both files are encrypted (AES-GCM) — only littlEye AI opens them and a changed byte is detected. A .le3AI carrying a model in .mlpackage format (chapter 13) requires version 3.1 or later.
- Compatibility: the .le3AI and .le3set formats are unchanged from version 3.x. A model or dataset from 3.x opens in 4.0 and a file from 4.0 also opens in 3.x (except .mlpackage models, which need 3.1 or later). Exporting a model keeps its original author.

Data provenance is part of the model

A model travels for years. Without a recorded source and licence nobody can later judge honestly what may be done with it — whether commercial use is allowed, whether the licence must be preserved, whom to cite. That is why .le3AI carries them and why the app requires them when importing a foreign model.

13 Importing a model trained outside the app

.mlpackage / .mlmodel, provenance, in-app test

littlEye AI also accepts a model trained outside the app — typically in Python (PyTorch) and converted with coremltools to .mlpackage or .mlmodel. Everything the app stands on is preserved: the model runs on-device, its provenance is recorded and the app measures the metrics itself.

1. Models → Add model → External Core ML model… (on the Mac also File → ⇧⌘O, from any section). Choose a .mlpackage or .mlmodel.
2. The app checks the model: it must have one image input and classifier outputs (class + probabilities). Otherwise it is rejected with a precise hint what to change in the conversion.
3. Map each model class to a catalogue diagnosis (prefilled when the class name matches) or type your own name. This makes the preliminary diagnosis and the in-app test work.
4. Fill in the data source and licence — required; without them the import does not complete.
5. After import press Test on my images…: the model runs on images from your dataset with the same diagnoses and the result (accuracy, confusion matrix) is recorded as its only metric. The app never shows numbers taken over from elsewhere.

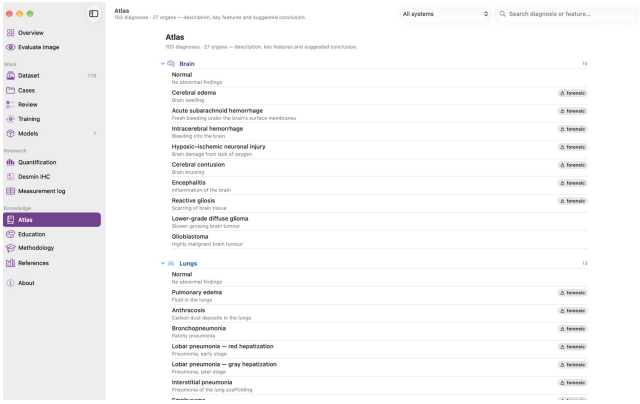
Conversion in Python (minimum)

```
import coremltools as ct
mlmodel = ct.convert(traced_model,
inputs=[ct.ImageType(name="image", shape=(1, 3, 224, 224), scale=1/255.0)],
classifier_config=ct.ClassifierConfig(["pluca.edem", "pluca.antrakoza"]),
minimum_deployment_target=ct.target.iOS17)
mlmodel.save("model.mlpackage")
```

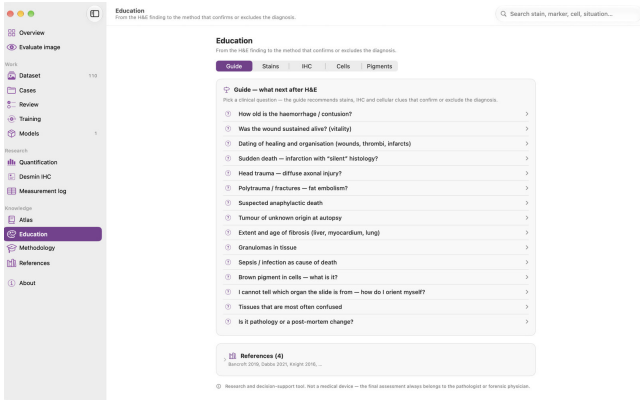
If you use littlEye AI catalogue IDs as class names (e.g. `pluca.edem`), the mapping fills itself in at import. Models over 200 MB are hard to send by mail; over 1 GB the app rejects them.

14 Atlas, Education, Methodology, References

103 diagnoses, stains, IHC, citations



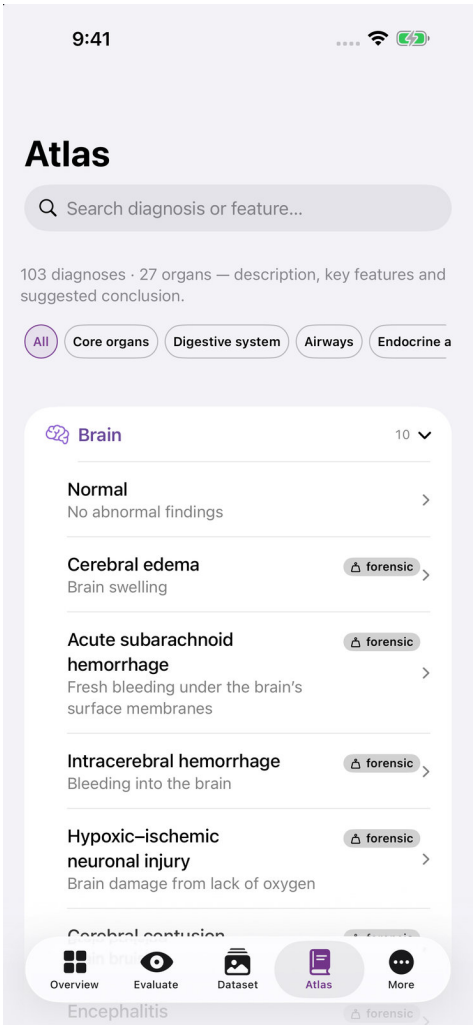
Diagnosis atlas — 103 diagnoses across 27 organs, search by name or feature, organ-system filter.



Education — the “what next after H&E” guide, stains, IHC, cells, pigments.

Diagnosis atlas

1. Search by diagnosis name, microscopic feature or conclusion; narrow the organ system with the picker. The search covers both Slovak and English.
2. Organs are collapsed into cards; organs that so far have only the normal pattern are merged into one section.
3. A diagnosis detail shows the microscopic appearance, key features, a suggested wording of the conclusion (Copy button), recommended next steps (special stains, IHC) and a Sources card with links.
4. On the Mac ⌥⌘A opens the Atlas in a separate window — you can keep it next to Review or Evaluate.



Education

The “what next after H&E” guide starts from a clinical question (dating a haemorrhage, wound vitality, silent infarct, DAI, fat embolism, anaphylaxis, tumour of unknown origin, autolysis vs. pathology…) and recommends stains, IHC markers and cellular clues that support or exclude the diagnosis. The categories (stains, IHC, cells, pigments) sit in one row under the title; an item's “Open tool” button leads straight to Quantification with the recipe preselected.

Methodology and capture

Methodology has two tabs. Model training explains “How the app works” in four steps, the capture rules and study ideas; Capture and hardware describes how to get an image from a microscope camera, scanner or UVC camera into the app — the app needs no drivers: microscope camera → export → Files/iCloud/AirDrop; USB-C UVC cameras work natively; slide scanners with TIFF/JPEG export.

References and sources

All medical content is cited — textbooks (Robbins & Cotran, Knight, DiMaio, Bancroft, Dabbs), the AECVP recommendation on myocardial infarction, papers on wound dating and β-APP, the WHO classification, PathologyOutlines and the NCT-CRC dataset. Links go to the DOI, a library catalogue or the official site. You see the source with each item; References in the sidebar holds the complete list with search, an area filter and a note on where in the app each source is used (e.g. Atlas › Heart).

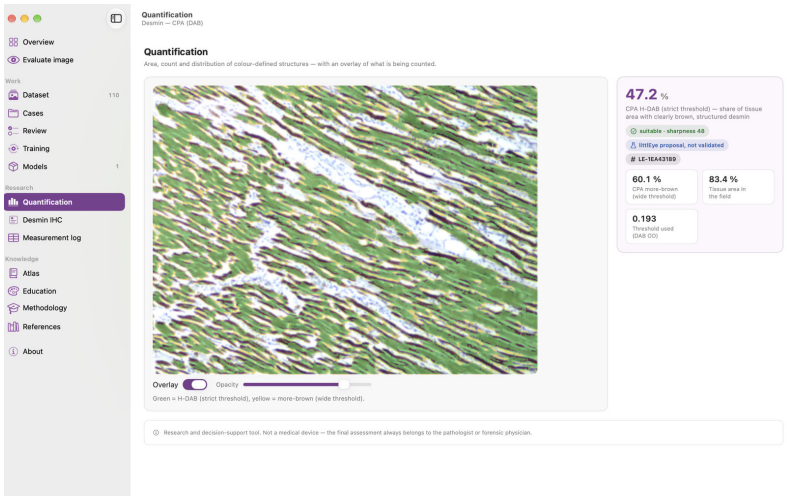
15 Image quantification and desmin IHC

instrument panel · recipes · study mode · CPA and indices

littlEye AI has two tools that use no trained model but measure colour and shape on the image directly: Image quantification (area, count, size and distribution of colour-defined structures) and Desmin IHC (a calculator of CPA and indices based on the author's dissertation). Both sit in the sidebar in the Research group (⇧⌘Q, ⇧⌘D), together with the Measurement log (chapter 16).

The instrument panel

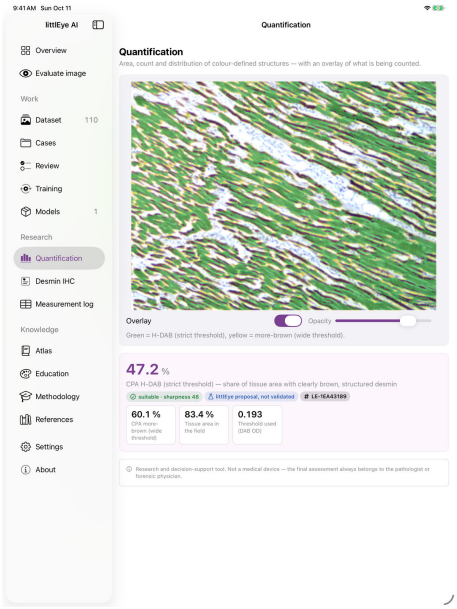
Both screens are built as a measuring instrument: the image with an overlay of what is being counted is on the left (on the iPhone at the top) and beside it a large result with its interval and a sentence on what the number means and what it does not. Below is the Settings and verification card — sensitivity (threshold), fields, point counting and sensitivity analysis. Until a result exists the settings are expanded; with a locked protocol they are collapsed and locked. Study mode (locked protocol, batch, blinding) appears as a status strip above the panel; the “Study…” button sets it up. The interval in the result is ± SD from the fields if they are computed, otherwise the 95 % interval from point counting, otherwise it is not shown.



Quantification on the Mac: the image with the overlay on the left, the result with its interval on the right, settings and verification below.



iPhone: image, result and the Settings and verification button.



iPad: in portrait the panel stacks, in landscape it sits side by side.

Image quantification

Pick a recipe (what to measure), load an image — from case slides, Photos, Files or by dropping it onto the canvas — and the result is immediate: a number, a colour overlay of what was just counted, and a sentence on what the number means and what it does not. The sensitivity slider and the overlay exist so that you always see what you are trusting. The image can be sent to Desmin IHC (Open in Desmin IHC → DAB / → H&E); the image is sent, not the numbers.

Recipes

- Lipofuscin (H&E) — area, count and size of yellow-brown granules, granules per 100 nuclei (a stand-in for “per cell”) and the perinuclear share. Colour alone does not separate lipofuscin from hemosiderin or formalin pigment — confirm with Perls, PAS, long Ziehl-Neelsen.
- Vacuoles / steatosis (H&E) — clear round spaces: area, count, macro- vs. microvesicular by nuclear diameter, size histogram; NASH CRN bands as orientation only (they refer to the share of hepatocytes, not to area).
- Erythrocytes (H&E) — area of bright red, extent across the field (diffuse vs. focal), the largest contiguous focus. Area does not separate congestion from hemorrhage — that is for the physician by location.
- Nuclei and PMNL (H&E) — count and density of hematoxylin nuclei; small dark lobulated ones as an orientational PMNL share. Tap correction: on an object = exclude, on empty space = add.
- Desmin — CPA (DAB) and Myocytes on H&E — the same measurement the Desmin IHC calculator uses. On H&E only strong pink (myocytes) is counted; pale pink (collagen, interstitium), nuclei and empty spaces are excluded and shown in a different colour.

> **Scale and units**
>
> Without a scale, sizes are in pixels and densities per field. Enter the field width in µm (measure with a stage micrometer at the same objective) and you get µm, µm² and counts per mm². The value is remembered.

What quantification is not

It measures colour and shape on a photograph — not a diagnosis. Thresholds are sensitive to exposure, white balance and section thickness; only images from the same protocol are comparable. For publication verify the values in ImageJ on a whole-section scan. Copy summary puts all numbers and settings on the clipboard.

Study mode

Quantification is a measuring instrument, so it does what a study asks of one:

- Locked protocol — you name the series (e.g. “Desmin-2026-A”) and the app locks the recipe, threshold, deconvolution and scale. While it is locked the sliders cannot move and every measurement records the protocol name. Without it you would be measuring the agreement of settings, not of the method.
- Blinded measurement — hides the image name and the diagnosis when picking from case slides; the rater does not know what they are measuring. The log entry is marked as blinded.
- Measurement log — the “Record in the log” button stores the measured values together with the date, the rater (the name from Settings), the protocol, the threshold setting, the scale and the app version. The same button is in the desmin calculator.
- Repeatability and agreement — measure the same image more than once and the log shows the mean, SD and coefficient of variation (CV). When several raters measure the same images, or one rater twice, the log also computes ICC(2,1) with a 95 % confidence interval and a Bland–Altman plot (bias, limits of agreement) — see below.
- CSV export — one row per measurement, semicolon separator and a decimal point; opens in Excel, R and Python without edits. It also carries the settings checksum, the staining batch with its factor and the normalised main value. ROC or regressions are computed from there.

Procedure for a series

1. Fill in the rater's name in Settings. 2. On a representative slide estimate the stain vectors, choose the threshold (Otsu or manual), check the overlay, run the sensitivity analysis and verify the threshold by point counting — the pixel value should lie within the interval. 3. Lock the protocol. 4. Record the batch control slide. 5. Turn on blinding. 6. Measure; every image is first checked by quality control; measure a whole-section scan across fields, then record. 7. Measure a subset a second time and with a second rater. 8. Export the CSV at the end and copy the Methods paragraph.

Tools for publication

The core of the measurement — deconvolution → threshold on the optical-density channel → positive-area fraction — is the workflow behind the community's reference tools (ImageJ Colour Deconvolution and IHC Profiler, QuPath); area is measured because DAB is not stoichiometric and intensity is unreliable (Taylor & Levenson 2006). On top of that the app adds what a reviewer asks about first — and what so far had to be done outside the app:

- Image quality control — every loaded image gets a badge suitable / borderline / unsuitable with the reason: sharpness (the no-reference blur metric of Crete et al. 2007), tissue share, clipped highlights and shadows, white point and colour cast of the background. A bad image thus does not become a data point nobody can tell from a good one.

- Threshold sensitivity analysis — a button scans the whole slider range and draws the curve of the value; the red line is your setting, the green band the plateau (change under 0.75 % per unit), the blue band the ±10 window. The result states whether the value is robust, moderately or highly threshold-sensitive, and the range can be reported in the paper. For Index 2 both thresholds (DAB and H&E) are scanned and combined into one range.
- Methods paragraph — ready text for Materials and Methods, in Slovak and English: app version, the deconvolution citation and stain vectors, optical-density thresholds (not slider positions), raster size, scale, quality control, sensitivity analysis, locked protocol and blinding. It ends with the settings checksum (LE-XXXXXXXX): a fingerprint of the version, recipe, thresholds, vectors and scale — the same checksum means the same setting. The checksum is also on every log entry and in the CSV.
- Stain vectors from your own slide — the standard Ruifrok & Johnston vectors were measured on other material; the Estimate from the image button derives them from your slide by the method of Macenko et al. (2009) and shows the deviation from the standard in degrees. Units are anchored to the standard optical density, so the threshold stays comparable. For a study: estimate once from a representative slide of the batch, lock, do not change.
- ICC and Bland–Altman in the log — when two raters measure the same images (between) or one rater twice (within), the log shows ICC(2,1) with a 95 % confidence interval (Shrout & Fleiss 1979; McGraw & Wong 1996), the Koo & Li 2016 band and a Bland–Altman plot with bias and limits of agreement. At least five shared images.
- Staining batches — CPA values are comparable only within one protocol, and the staining batch is its most volatile part. Stain the same control slide in every batch, measure it with the locked protocol and record it as the batch control slide. The log relates every batch to the first (factor = reference control / batch control), draws the drift and warns above 15 %; the CSV carries the factor and the normalised value for every measurement.
- Automatic threshold (Otsu) — instead of the slider, the optical-density histogram of the tissue sets the threshold (Otsu 1979), with no rater input; the OD value used is in the result and in the Methods text. Otsu tends to be stricter than a hand-calibrated threshold: on the calibration image it gave 30 % against 47 %. Neither is “the truth” — point counting decides; then lock the chosen method in the protocol.
- Distribution across fields — a scan or large photograph of the section is divided into an 8 × n grid; every field is measured with the same recipe and the same fixed threshold (systematic sampling, Gundersen & Jensen 1987). Result: mean ± SD, CV, range, number of fields and a map (red = low, green = high). For desmin it decides between focal loss (some fields below half the median) and a diffusely low value. It is also in each slot of the image pair.
- Point counting — stereological check — a grid of ~250 points with a random offset (repeatable seed) is laid over the image; points on glass are discarded, every point gets its class from the measurement and you reclassify it by tapping. The point fraction is an unbiased estimate of the area fraction with a Wilson 95 % CI — classic morphometry against which the automatic figure is verified. The app states at once whether the pixel value lies within the interval.
- DAB + H&E pair check — for Index 2 the app compares the format and size of both images, the tissue share and, as orientation, the nucleus size, and states whether the pair is comparable. Magnification cannot be verified reliably from the content — certainty comes from the same objective and camera setting for both images.

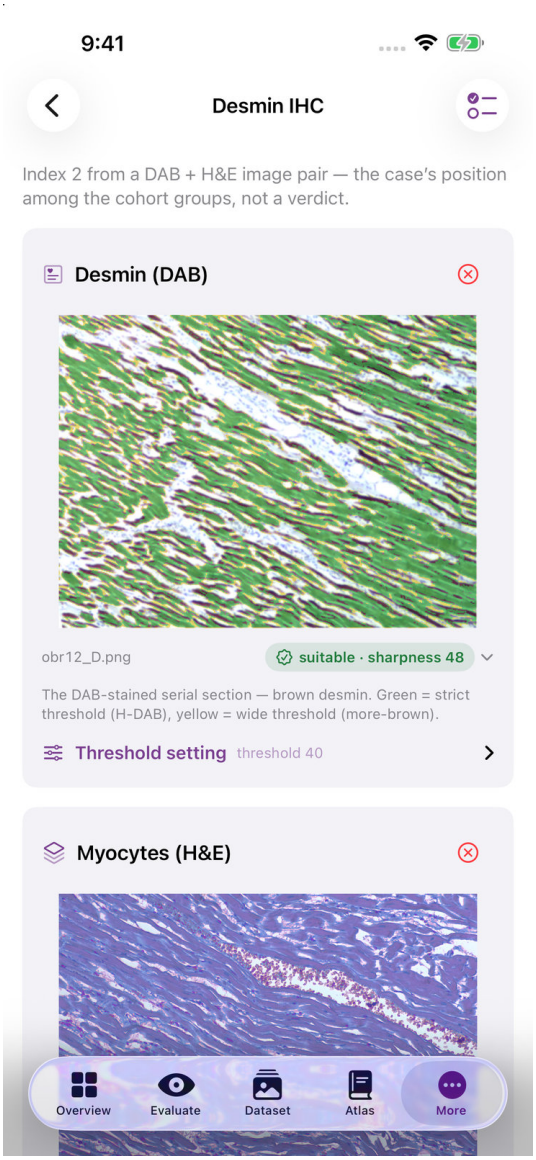
Study ideas are in the app itself (Study mode → Study ideas and collaboration): the question, the design and the pitfall for five topics feasible with these tools at a single institution — from comparing Index 2 with a bare CPA, through skeletal muscle as a control of post-mortem change, to a methodological paper on reproducibility.

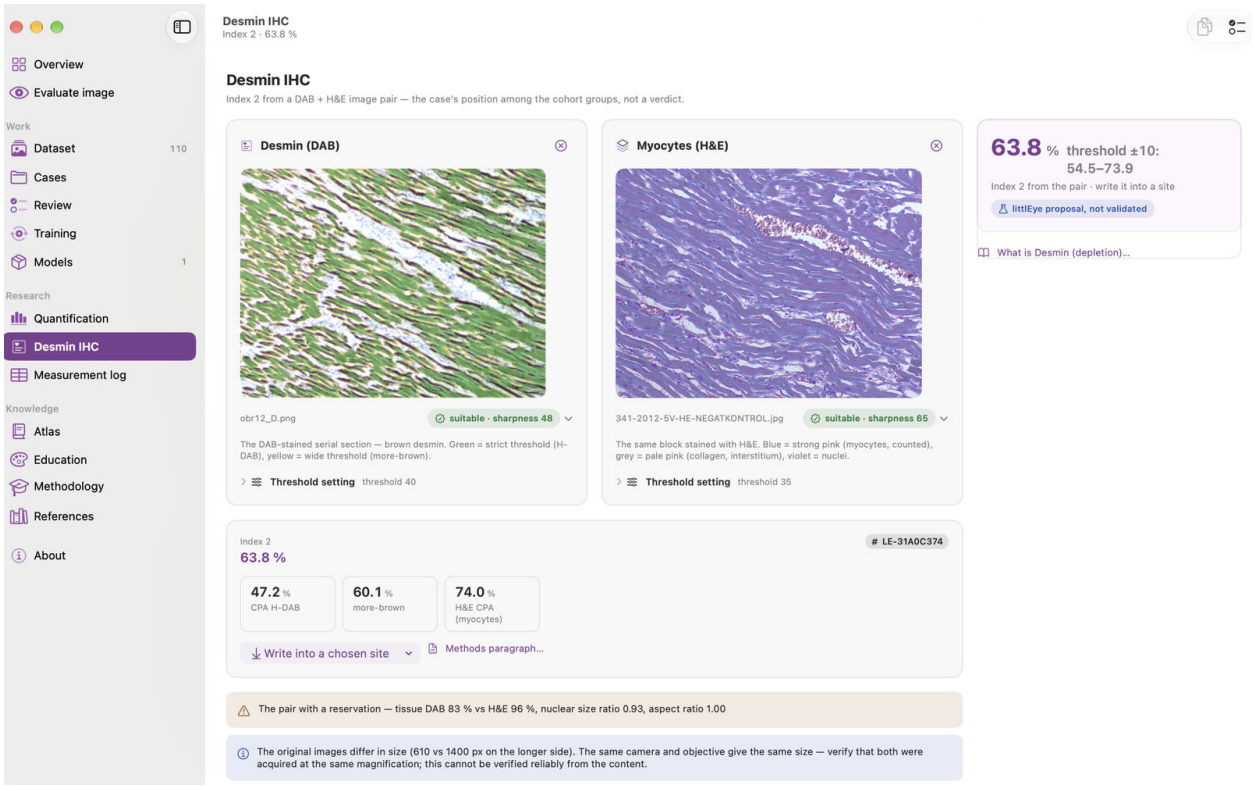
Desmin IHC — CPA and indices

Desmin is a structural protein of cardiomyocytes that is lost in ischaemia and in agonal hypoxia alike — the loss of staining is the finding. The author's dissertation (Šikuta 2021, n = 89) quantified the extent of this loss as CPA (Colour Proportionate Area, % of area with brown DAB) and found that the CPA in confirmed infarction (12.4 ± 7.9 %) and in “negative” controls (12.8 ± 5.4 %) overlapped: depletion accompanies any death with an agonal phase. The calculator therefore gives no verdict — it gives the position among the cohort, the pattern and an autolysis control. The primary measure is Index 2 from a pair of images of the same block.

Since 4.0 Desmin is a single screen without an Evaluate button: the result is recalculated live as soon as you change an image or a value.

1. At the top are two Image pair slots: load into the first a scan or microphotograph of the serial section stained with DAB, into the second a photo of the same block stained with H&E (same magnification). A slot also accepts an image sent from Quantification. Each slot shows the overlay, a quality badge and a collapsed threshold setting.
2. Below the slots you see Index 2, the pair comparability check and the Write to the chosen site button. Writing is always a manual confirmation — nothing is added by itself. For another sampling site load a new pair. Values can also be typed in by hand (from ImageJ) in the Sampling sites card.
3. Enter the sites separately — infarction is focal and a mean dilutes it. At least three sites are recommended.
4. In the Confounders and controls card fill in the death → cooling interval (above 12 h the CPA falls by itself, markedly from 32 h), pancreatic autolysis, resuscitation and desmin in the vascular smooth muscle of the same section — the cheapest control: preserved in vessels + lost in myocytes = the loss is myocyte-specific; lost in both = autolysis or a staining failure.
5. The result appears on the right: Index 2 with its interval, the pattern (focal loss / diffusely low / even), the position among the cohort (a chart with four groups), the indices and a list of what could have lowered the number without ischaemia. Right beside the main value and the pattern stands the badge “littlEye proposal, not validated”.





Desmin IHC on the Mac: the DAB + H&E pair on the left, Index 2 and the position among the cohort on the right, sampling sites and confounders below.

The indices — methodology

- Index 2 = H-DAB CPA / H&E CPA × 100 — the primary measure. Both figures are shares of the same section area from a pair of images of the same block, so the ratio states directly what share of the myocyte area is desmin-positive. The H&E denominator excludes empty spaces, interstitial oedema, nuclei and pale collagen (only strong pink counts), so the index does not depend on how much fibrosis and oedema the section holds — unlike a bare CPA. The natural ceiling is 100 %.
- Index 3 = heart CPA / skeletal-muscle CPA × 100 — the autolysis control. Skeletal muscle from the same body (diaphragm, psoas; the same staining batch) shares the post-mortem interval, fixation and technique, but not the agonal ischaemia of the myocardium. If the heart is markedly below skeletal muscle, post-mortem change cannot account for the difference — the loss is selective for the myocardium. If they are comparable, Index 3 alone does not decide; Index 1 and the interval complete it. The index is deliberately relative — it has no absolute threshold, so it works in a differently calibrated laboratory too.
- Index 1 = H-DAB CPA / more-brown CPA × 100 — staining preservation. The study's original hypothesis (lower index = more ischaemia) failed, but its own data show what the index measures: a short post-mortem interval ≈ 70 %, from 32 h ≈ 50 %, putrefaction 6–27 %. It is therefore a measure of degradation of desmin, not of ischaemia — and in that role it is useful.

The autolysis control

From Index 1, Index 3, vascular desmin, the post-mortem interval and pancreatic autolysis the app composes a single sentence: autolysis as the explanation is unlikely / cannot be excluded / is likely — and lists which data led there. The purpose is one: to separate “desmin is missing because the myocardium was ischaemic” from “desmin is missing because the body was decomposing or the staining failed”. The weighting is a littlEye proposal, not validated.

How to read the result

There is no deciding threshold. Informative is a focal loss with sharp demarcation in a well-preserved body; diffuse loss is non-specific (agony, resuscitation, asphyxia, post-mortem interval, putrefaction). The cohort numbers are descriptive data from one institution and protocol, not reference bands. To confirm early ischaemia, C5b-9(m), fibronectin and fibrinogen are better established — they accumulate in ischaemia rather than disappear. The final assessment belongs to the physician.

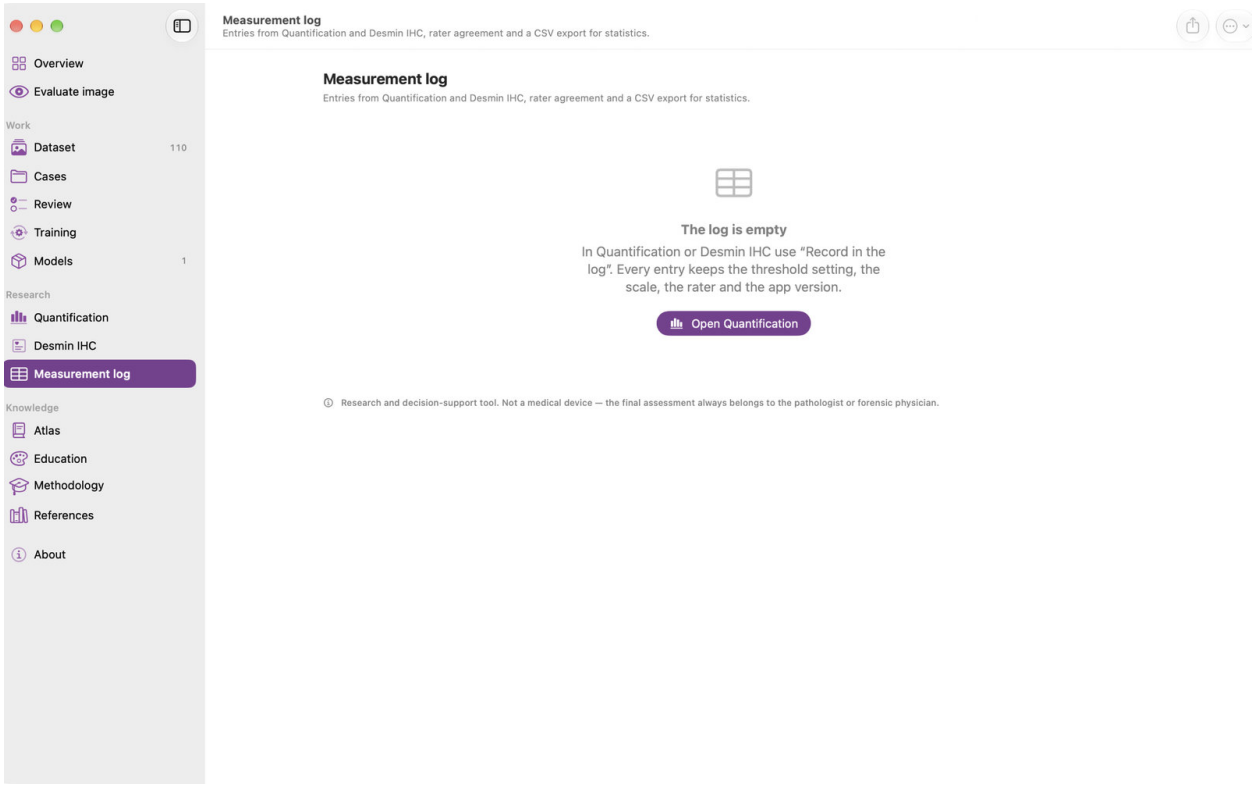
Where the CPA comes from

The in-app measurement is built for a study: locked protocol, own stain vectors, quality control, sensitivity analysis, batch control slide and a Methods paragraph with a checksum. ImageJ with the IHC Toolbox plugin (H-DAB and H-DAB more brown models) on a whole-section scan remains the independent cross-check — compare both paths on a subset of cases and report the agreement.

16 The measurement log

table, batches, ICC and Bland–Altman, CSV export

The Measurement log is its own section (⇧⌘L) in the Research group. It collects entries from Quantification and Desmin IHC, computes rater agreement and prepares the export for statistics.



The Measurement log on the Mac: the table of entries with sorting, staining batches and rater agreement.

- An entry is created by the "Record in the log" button in Quantification or in Desmin. It stores the date, the rater (the name from Settings), the protocol, the threshold setting, the scale, the settings checksum, the batch and the app version; blinded entries are marked.
- The table (Mac): date, image, recipe, value, protocol, rater and batch. Click a column header to sort; a right-click or ⌫ on selected entries offers Copy CSV row, Open in Quantification / Desmin and Delete entry… (with confirmation; a deleted entry cannot be restored, so export the CSV first). On iPhone and iPad it is a list with swipe to delete.
- Export CSV… in the toolbar: semicolon separator and a decimal point; it opens in Excel, R and Python. On the Mac via Save As, on iOS via sharing. The file carries all settings, the checksum, the batch with its factor and the normalised value. The menu also holds Clear the log….
- Repeatability: if you measure the same image more than once, the log shows the mean, SD and coefficient of variation (CV); below about 5 % the measurement is stable at the given setting.
- Rater and repeat agreement: if several raters measure the same images, or one rater twice, ICC(2,1) with a 95 % confidence interval is computed and a Bland–Altman plot (bias, limits of agreement) drawn. At least five shared images are needed.
- Staining batches: a control slide measured in every batch with the same protocol. The log relates every batch to the first (factor = reference control / batch control), draws the drift and warns above the 15 % tolerance. You create a batch in the Study sheet and can delete it in the log; the measurement entries stay.

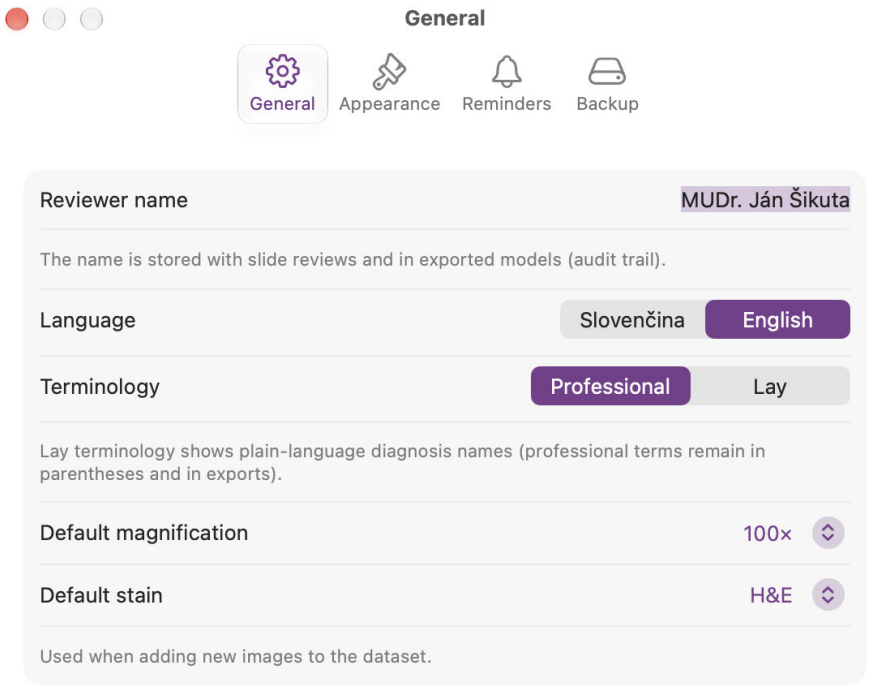
Backup and the log

The log and the batches are part of the complete library backup (chapter 17). If part of the log cannot be read, the app sets the original file aside and shows the readable entries.

17 Settings, backup and restore

four tabs · complete backup · restore from backup

On the Mac you open Settings with ⌘, (or from the app menu) as a separate window with four tabs. On the iPad it is a sidebar row, on the iPhone you find Settings under More. The content is the same.



Settings on the Mac: the General, Appearance, Reminders and Backup tabs.

- General: the reviewer name (stored with reviews and in exported models), the SK/EN language, Professional/Lay terminology and the default magnification and stain for new images.
- Appearance: the app icon — a grid of previews; on the Mac the Dock icon changes while the app is running. The app supports light and dark mode.
- Reminders: the switches “Show the collaboration reminder on Overview” and “Show the support reminder on Overview” and the buttons Collaboration and validation… and Support the project…. Reminders appear only on the Overview, never above your work.
- Backup: Back up library… and Restore from backup… (details below).

Backup and restore

1. Back up library… saves the whole library — images, cases, models, the measurement log and the batches — into a folder you choose (for example an external drive, USB-C or SMB). Nothing is sent anywhere. A backup is always complete, never incremental. The same command is in the File menu.
2. Restore from backup…: you choose the backup folder and decide. Merge into the library adds what is missing and changes nothing existing. Replace the library with the backup moves the current library into a “pred-obnovou-…” (before restore) folder — nothing is deleted.
3. While an import is running a restore cannot be started; the app says so. For an incomplete backup it reports how many files could not be copied.

Why back up

Everything is on the device, nothing in the cloud — and so also nothing that could be restored from a server. A regular backup to disk is the only safety net. A damaged archive index is also no longer overwritten: the app offers to restore the previous version.

18 littlEye AI on the Mac

menus, shortcuts, windows, drag & drop

On an Apple silicon Mac littlEye AI is a native app with a menu bar, keyboard shortcuts and multiple windows. Training on the Mac uses the whole chip and is fastest — the Mac is therefore the department's ideal “training machine”, while iPad and iPhone collect and review.

Menus

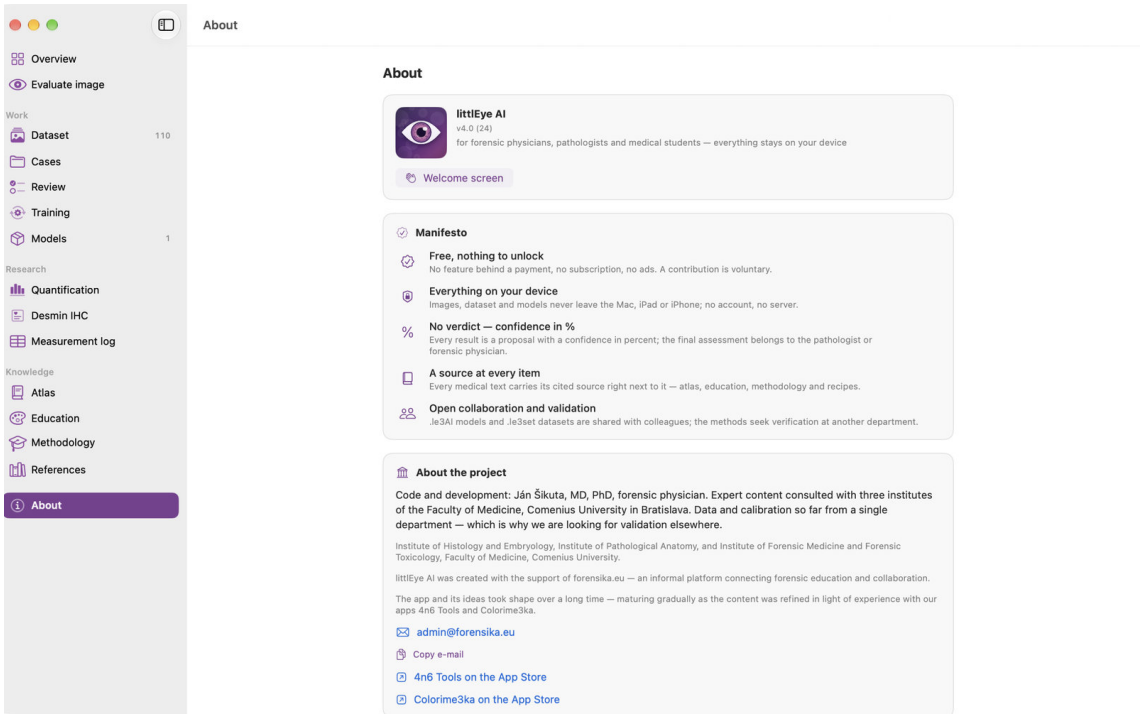
- File: Import model or dataset (⌘O), Import external Core ML model (⇧⌘O), Import images (⇧⌘I; active in Dataset, an organ and a case), New case (⇧⌘N), Export model (⇧⌘E; when a model is selected), Back up library and Restore from backup.
- Go: the same sections, in the same order and with the same shortcuts as the sidebar.
- Review: this menu appears only during review — Next / Previous slide (arrows), Skip (Space), Undo last vote and diagnoses 1–9.
- Window: Atlas in a new window (⌥⌘A). Help: the guide (PDF), website, Collaboration and validation and Support the project.
- About littlEye AI is a native panel with the author, institutes, contact and the notice; you can open it even without an open window.

Shortcuts (also work on iPad with a keyboard)

⌘1 … ⌘5	Overview · Cases · Evaluate image · Training · Models
⌘6 … ⌘9, ⌘0	Atlas · Education · Methodology · References · About
⇧⌘A	Dataset
⇧⌘H	Review
⇧⌘L	Measurement log
⇧⌘Q	Quantification
⇧⌘D	Desmin IHC
⇧⌘I	Import images
⇧⌘N	New case
⇧⌘E	Export model
⌘O · ⇧⌘O	Import model/dataset · external Core ML model
⌥⌘A	Atlas in a new window
⌘,	Settings
1 … 9 · Space · ← →	Review: diagnosis · skip · browse
⌘Z	Review: undo the last vote
⌘+ · ⌘– · ⌥⌘0	Zoom in · out · actual size of the image
Esc	Closes the open sheet

Windows and drag & drop

- Slide details open in a separate window (double-click or right-click → Open in a separate window) — you can keep it next to the case. The Atlas has its own window (⌥⌘A).
- A .le3AI or .le3set dropped onto the window is imported; images dragged onto an organ in Dataset, into a case or into Evaluate image are loaded. Images dropped outside an organ first ask for the organ.
- A right-click offers a context menu in Models, Cases, Dataset, the Measurement log and the Desmin slots; deleting always asks for confirmation. Space in the image grid opens a quick look.
- Training progress also shows as a percentage on the Dock icon. Files (.le3AI, .le3set, CSV, backup) are saved via Save As.



About: manifesto, about the project, collaboration and validation, support the project, privacy and model provenance.

19 Privacy, limits, collaboration and support

what the app does not do · collaboration · voluntary contribution

What the app does not do

- Does not connect to the network (exceptions: the App Store for a contribution, literature links in the browser). No accounts, no telemetry.
- Does not give a diagnosis. The model output is always only a suggestion with a confidence percentage.
- Contains no real slides and no ready-made diagnostic models — only the synthetic DEMO is bundled. You train models yourself or exchange them as files.
- Does not fine-tune models — it retrains them from scratch on a larger dataset (chapter 10).

Limits worth knowing

- The model only ever chooses among the classes it learned. A model of normal tissue will not recognise pneumonia — it labels it as the closest thing it knows.
- If normal and finding come from different image sources, the model reads the scanner's look instead of the tissue; a negative result then excludes nothing. Collect both classes from the same source.
- Before using a model in practice: validation on an independent set, sensitivity/specificity, agreement with a pathologist.

Collaboration

littlEye AI comes from a single institution. For the models and quantification methods to mean anything, they need validation on someone else's material — another scanner, another laboratory, other hands. We are looking for institutions that can offer H&E slides with a clear licence to share, validation of models and recipes on their own images, or co-authorship of methodological publications (desmin CPA, image analysis). We offer co-authorship, credit in the app and models free of charge. The offer appears twice in the Project card on the Overview and permanently in Settings → Reminders and About; write to admin@forensika.eu.

Everything is together in the Collaboration and validation sheet (the Validation and collaboration proposal button on the collaboration card, on the Mac also Help → Collaboration and validation…): using the app in your department, a ready multi-centre validation-study proposal for desmin quantification (material, a common locked protocol verified by the settings checksum, a batch control slide, blinded measurement by two raters with repeats, measurement across fields and point counting, CSV export — images are not sent), the collaboration offer and project support. The proposal can be copied as text into an e-mail or a grant application; the We'd like to join the validation button opens a pre-filled e-mail.

Supporting the project

littlEye AI is developed and maintained by a single physician as a spare-time project. If it helps you, you can contribute under About → Support the project — a coffee, a lunch, whatever suits, repeatedly if you wish. The App Store does not allow arbitrary amounts, hence fixed levels. Contributions fund upkeep and further development and unlock nothing. The reminder in the Project card on the Overview goes quiet after a contribution.

Contact and links

Website and guides: dzanino.github.io/littlEyeAI · Models on request and questions: admin@forensika.eu · The Forensika project: forensika.eu

littlEye AI 4.0 (24) · October 2026 · Ján Šikuta, MD, PhD, Institute of Forensic Medicine, Comenius University Bratislava. Developed in collaboration by specialists from three institutes of the Faculty of Medicine.